

- ① Apply the rules and step by step create a Turing Machine to accept $L = \{a^n b^n\}$ (7 marks)
- ② Write down 7-tuple definition for the Turing machine. Construct the Turing machine and its transition table over $\Sigma = \{a, b\}$ for the language $L = \{a^n b^n \mid n \geq 1\}$ (7 marks)
- ③ Draw Turing machine that accepts the language $\{a^n b^n c^n \mid n \geq 0\}$ over $\{a, b, c\}^*$. Also trace out the same on the input string aabbcc (7 marks)
- ④ Draw only the transition table of TM to accept the language $L = \{0^n 1^n : \text{where } n \geq 1\}$ (4 marks)
- ⑤ Develop a Turing Machine to accept the language $L = \{x \mid N_a(x) = N_b(x), x \in \{a, b\}^*\}$ (7 marks)
- ⑥ Apply the rules and step by step create a Turing Machine to accept $\{a, b\}^* \{aba\}$ (7 marks)
- ⑦ Draw a TM to accept Palindromes over $\{a, b\}$ (Even as well as odd Palindromes). (7 marks)
(w21, s21, w23, s24)
- ⑧ Write a TM to copy strings. (7 marks)
- ⑨ Construct a TM to accept the strings $x \cdot x^{\text{rev}}$ (7 marks)
- ⑩ Design a TM to delete a symbol. (7 marks)
- ⑪ Design a TM with no more than three states that accepts the language $a(a+b)^*$. (7 marks)
Assume $\Sigma = \{a, b\}$

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(7 marks)

For the given Turing Machine in Table-2, trace the transition for the strings 1011 and 10101 and identify the language recognized by this TM. TM is defined as $TM = (Q, \Sigma, \Gamma, q_0, \delta)$ where $\{q_0, q_1, q_2, q_3, q_4, q_5, q_6\} \in Q$, $\Sigma = \{0, 1\}$, $\{0, 1, X, Y, B\} \in \Gamma$, $q_0 \in Q$, $B \in \Gamma$, $B \notin \Sigma$, $\{q_6\}$ is the accepting state. (7M, S23)

Table-2

State	0	1	X	Y	B
q_0	(q_1, X, R)	(q_2, Y, R)	(q_6, X, R)	(q_6, Y, R)	(q_6, B, R)
q_1	$(q_1, 0, R)$	$(q_1, 1, R)$	(q_3, X, L)	(q_3, Y, L)	(q_3, B, L)
q_2	$(q_2, 0, R)$	$(q_2, 1, R)$	(q_4, X, L)	(q_4, Y, L)	(q_4, B, L)
q_3	(q_5, X, L)	—	(q_6, X, R)	(q_6, Y, R)	—
q_4	—	(q_5, Y, L)	(q_6, X, R)	(q_6, Y, R)	—
q_5	$(q_5, 0, L)$	$(q_5, 1, L)$	(q_0, X, R)	(q_0, Y, R)	—

- 13 Show that a TM to recognize the language $L = L(0^*1)$ can accept the string without moving the head in L direction (4 marks)

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- 1 Write a short note on Universal Turing Machine (4 marks)
- 2 write a short note on church-turing thesis. (4 & 7 marks)
- 3 Define and explain the working of TMs (3 & 7 marks)
- ↳ what is TM?
 - ↳ Explain its capabilities.
 - ↳ write advantages of TM over FSM